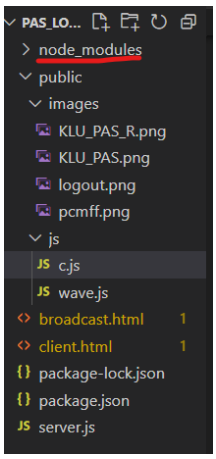
Steps to host PAS in Local server

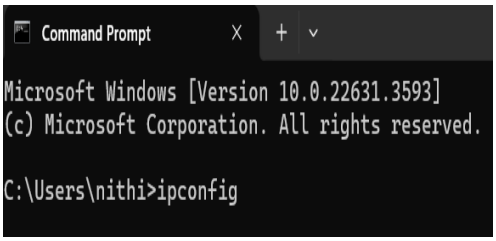
1) Open the folder with name “pas_local” in Visual Studio Code.
2) Install all required packages with the following command.This will add new folder in the directory called node_modules.
   “npm install node ws express”

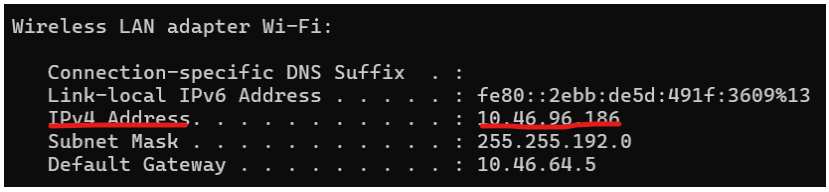


3) A) If you want to access only the website you can connect to any WiFi credentials but if you want to connect the website to the 32bit controller speaker you should strictly follow the 3(b) step or else ignore the step.

   B) Connect to a wifi which is not under captiveportal. In our case we have the router with the following credentials.
   SSID: RND
   Password: 123123456789

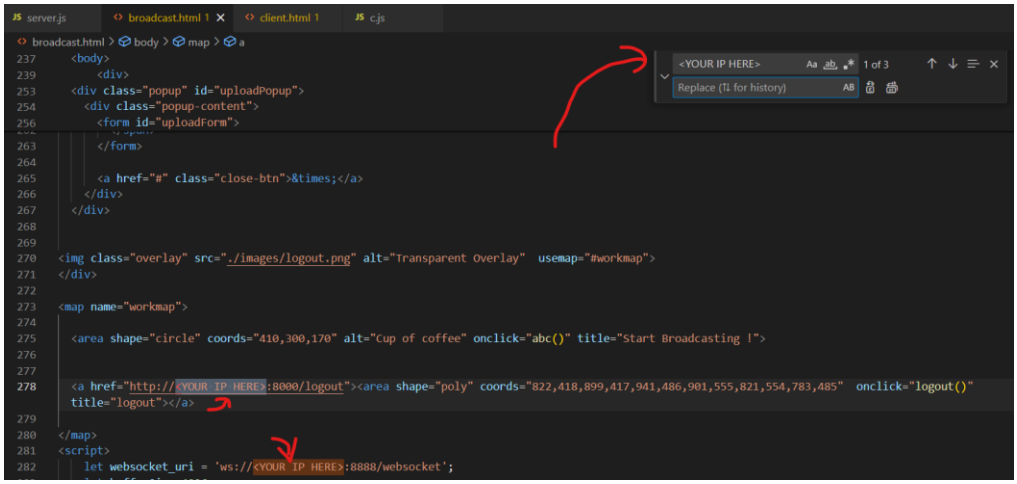
4) Open Command prompt , type “ipconfig” .Here copy the IPv4 address.



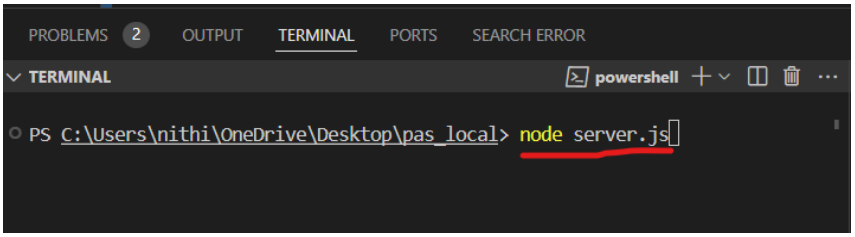


5) Now come back to VS code , here we need to make changes in the following three files which are broadcast.html , server.js , client.html.

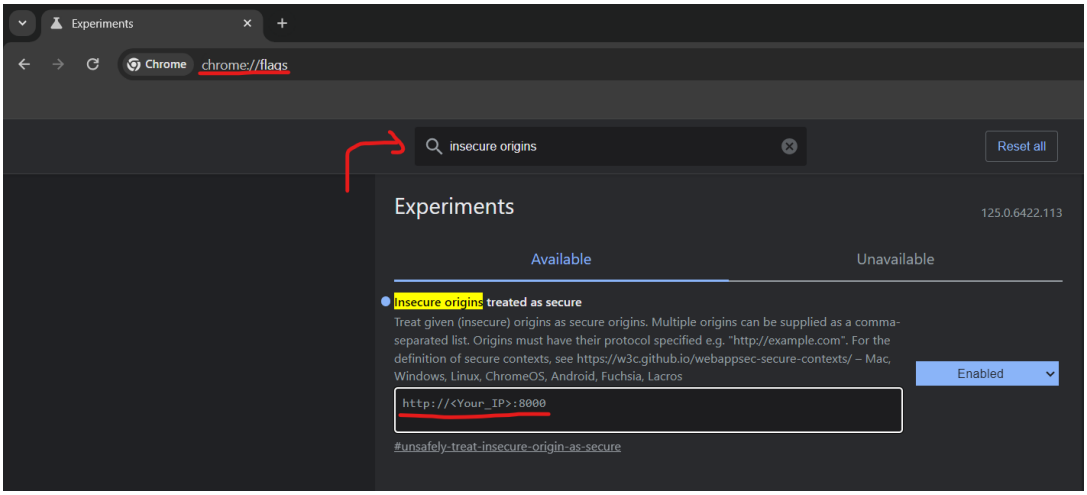
6) In all the three files where ever you find this “<YOUR IP address>”,replace it with the your IP address copied from the command prompt.



7) Go to new terminal in VS code and enter the command “node server.js”,this will result in the following output.



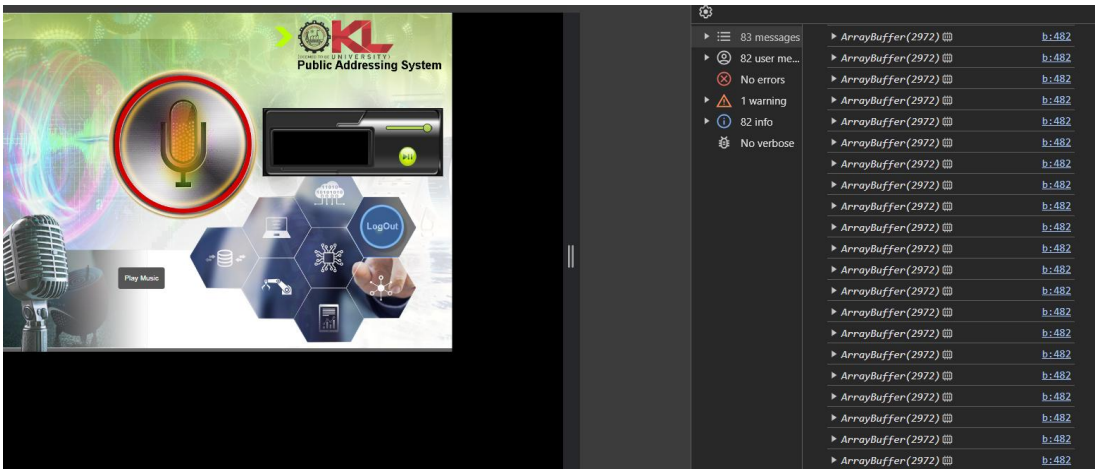
8) Open the following link in your browser “chrome://flags/”.In this search for “Insecure origins”.



9) Enter the following url in the text field http://<Your_IP>:8000 and click on Relaunch button.

10) Now you can access the perfectly working website, with the following URLs.
http://<Your_IP>:8000/b - broadcast.html
http://<Your_IP>:8000/c - client.html

11) Check the functionalities of the website such as broadcasting , playing music ,listening through the player in the same page.If any issue check error in the inspect window.

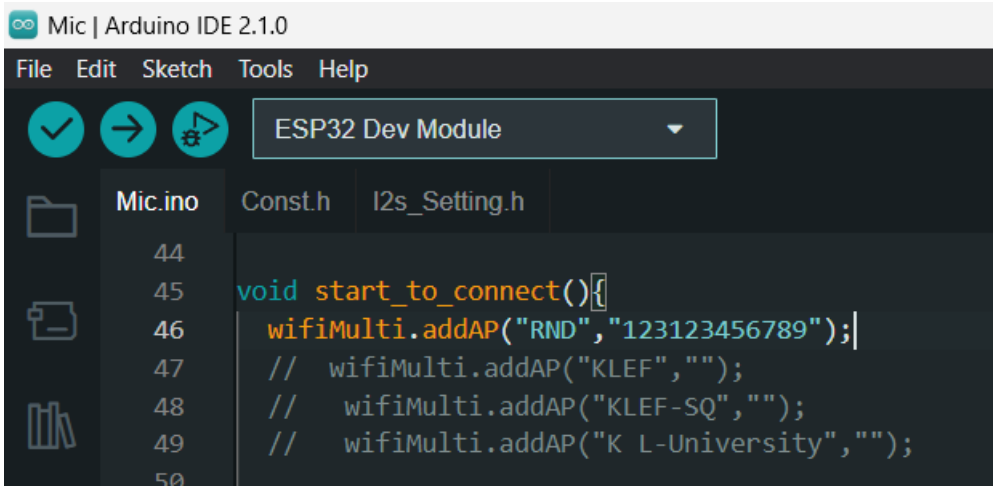


12) This makes the website setup successfully in the local server.Now the task is to setup 32bit controller speaker with our websocket.

13) In our case the 32bit controller we used is ESP-32,we have a total of three files in this code. But we need to make changes in the following two files which are Mic.ino and const.h.

14) In const.h we need to change the websocket URL.

15) Make sure you connect the ESP32 to the same network in which our laptop is connected by changing the credentials in Mic.ino file.

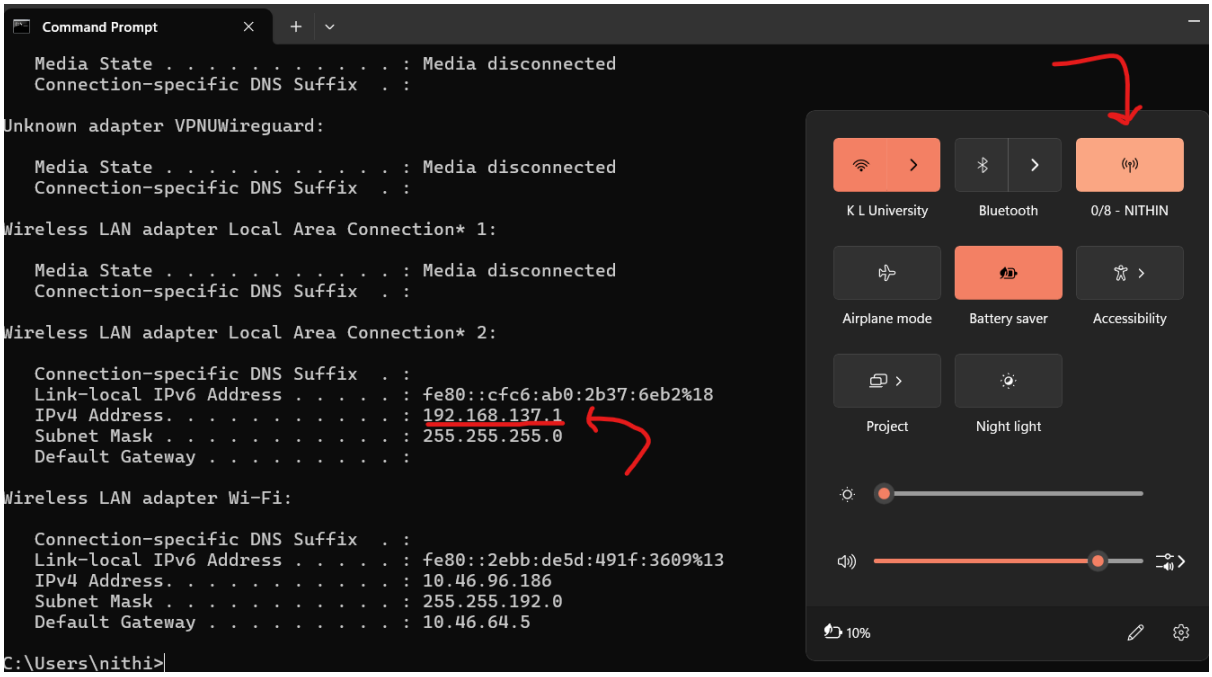


16) Upload the code to ESP32 and check in the serial monitor whether the ESP got connected with the wifi and the websockets.

17) These steps will lead to make a connection between 32bit controller speaker and the locally hosted website.

18) Check the output from the speakers by broadcasting from the website.

19) Go through the next page for basic instructions.

NOTE:

- Read the 3rd step throughlly.
- Keep an eye on the red marker in all images attached in this document.
- Make sure you change the IP in the code from time to time accordingly.
- Before setting up the speaker locally check whether the speakers are working in the global host or not.This helps in debugging the issue.
- You might face issue in the 17th step.Try to debug the issue in all the possible ways such as check as checking SSID , websocket_url etc:.
- **If everything is right still not working ,then only perform the next step.**
- Then try to connect the esp32 with your laptop hotspot credentials.Since your laptop hotspot is ON ,your IP will differ from previous.So,after turning on the hotspot make sure you follow the steps again from **Step3(b) to Step17.** With hotspot turned on your IP will be in the format of 192.137. * . *.